

Individual Report — Junho Hwang

CS453 Team 6

Student ID: 20250813

Name: Junho Hwang

Details of My Contribution

In this project, I was responsible for building a modular Python experiment automation pipeline to systematically evaluate the Diversity-Aware Mutation Fitness algorithm. The core functions of each Python file within the built automation package (automation/) are as follows.

config.py: Designed configuration code to define and manage the experiment directory path, output metric variables, hyperparameter sweep settings, and benchmark target specifications.

runner.py: Built a subprocess-based EvoSuite execution runner to automate timeout and error handling, and implemented a mechanism to automatically aggregate post-execution data from statistics.csv.

builder.py: Implemented logic to perform Maven compilation and directly examine the StrongMutationSuiteFitness.class bytecode within the generated JAR file to verify that the diversity measurement marker (MutantTypeEntropy) added to EvoSuite is correctly included.

env_manager.py: Wrote a module to verify the integrity of the experiment environment before execution by checking the Java/Maven environment, the status of the Git repository, and the existence of required dependencies for the benchmark class.

Peer Assessment of Team Members

Minwook Lee (minux-lee): (10/10)

Established the overall research direction of the project. Laid the academic foundation for the project by proposing a diversity metric based on Shannon Entropy. Gaining a deep understanding of the overall stack of EvoSuite, he successfully implemented a mutation diversity index tailored to the research objectives and applied it directly to the fitness function within the genetic algorithm. He clearly defined the specific phases and data required for hypothesis testing and performed statistical analysis based on the collected data. He led all meetings, presented clear tasks to team members, spearheaded the writing of the final team report, and took full responsibility for the final presentation, thereby leading the project.

Minwoo Kim (aijk679): (10/10)

He played a key role in the planning and execution of a large-scale experimental campaign. Based on the initial pipeline framework, he established the specific experimental parameters required for phases B, C, and D and implemented practical parameter sweep scripts. Additionally, he successfully executed thousands of automated EvoSuite run loops to collect reliable empirical data. Beyond his engineering contributions, he aided in establishing evaluation strategies by providing valuable theoretical insights into mutation testing concepts during the initial design phase.

Seonwoo Ji (neutrino1359): (10/10)

He laid the conceptual foundation for the project by first proposing the core idea of evolutionary testing and introducing EvoSuite. He also led the post-experiment phase, including developing data analysis scripts, managing comprehensive raw data collection, and calculating key performance metrics. He demonstrated outstanding research leadership by exploring and validating various exploratory fitness function formulas to derive meaningful trends. He produced key figures, graphs, and visual data necessary for the presentation materials.